SG5 Drowsiness Decision and Alert Logic

Midterm technical report and integration handoff

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G1-D2 Driver Drowsiness Monitoring | 28 October 2026

Executive summary

SG5 now has a reproducible, headless decision module connected to the available subgroup code. The PC pipeline runs SG1 face landmarks, SG2's selected EAR processor, a causal SG3 mouth adapter, the actual SG4 temporal engine, SG5 decision logic and the SG6 status-aware consumer. Five complete saved-video runs processed **2932 frames and 1173 temporal vectors**. Those clips have no independent drowsiness labels; their outputs establish interoperation, not detection accuracy.

The provisional primary is the robust C state machine; threshold A is the transparent backup. Five configurations were compared on the same 5,400 synthetic rows. Weighted B has the best overall sample accuracy but misses the short labelled episode. Robust C catches both episodes and fixes quality, timing and session handling; it is not the accuracy winner. The sensitivity candidate is retained for analysis rather than selected as a held-out winner.

The deliverable is a **PC-verified integration and SG6 software handoff**. SG3 calibration, SG4 adapter semantics and the proposed SG6 v2 fault policy need owner acceptance. Jetson boot, live camera delivery, GPIO operation and target performance are not verified here.

Evidence boundary	Status
Connected source-to-request pipeline and reproducible comparisons	Verified on PC
New adapter semantics and SG6 v2 contract	Owner review pending
Real-driver accuracy and generalization	Not established
Physical embedded operation	SG6 target checks pending

Reading map

2 Requirements and assessment evidence 3 Connected architecture 4 Producer contracts 5 Decision logic 6 Controlled results 7 Failure analysis 8 Real-video integration 9 SG6 handoff 10 Reproduction and tests 11 GitHub and embedded readiness 12 Remaining work and references

2 Requirements and assessment evidence

The current Stage 1 assessment is **A 15 + B 10 + C 5 = 30 marks** [1]. It is not the earlier Week 5 ten-mark subgroup rubric. The midterm permits a development-PC prototype and does not require every algorithm to be deployed on Jetson. This matrix maps evidence to the rubric; it does not assign marks.

Area	Marks	Demonstration evidence	Qualification
A	7	Working prototype: real SG4 replay, a complete saved-video pipeline, observable state/status and SG6 requests.	PC verified
A	4	Primary C and backup A; preserved B alternative; common-data tables, failure analysis and recovery sensitivity study.	Evidence recorded
A	2	SG4 v1.1 schema/golden parity; exact valid-only SG6 legacy adapter; explicit proposed v2 boundary and run commands.	Technical checks pass; new semantics pending
A	2	Unit/contract tests, strict metrics, no-face and silence tests; synthetic and unlabelled video limitations separated.	PC verified
B	3	Shared module structure; source snapshots and hashes; SG5 feature branch with separate integration additions.	Review and merge are separate
B	3	Existing private Project 2; assigned SG5 issues 8 and 9 plus midterm evidence/acceptance tracking.	Team board evidence
B	2	Module and SG6 READMEs, pinned PC dependencies, command sequence, examples, report and result links.	Reproducible on PC
B	2	Preserved 27 October SG4 alignment note, technical coordination register, owners and unresolved acceptance actions.	No invented meeting or approvals
C	2	Actual Jetson boot, storage and OS evidence must be shown by the team/SG6.	Not verified by this work
C	1	PC environment is pinned; SG6's reported detector-only target benchmark is separately attributed.	Target stack needs confirmation
C	2	Saved-video software path is tested; USB/CSI camera connection and target acquisition need SG6 evidence.	Live target capture pending

Delivery scope

SG5 owns temporal-cue fusion and alert decisions, not the authorship of imported subgroup algorithms. The guide places multi-module integration at Lab 8, individual target deployment at Lab 9 and a complete embedded pipeline at Lab 11 [2]. The report supports the present gate and records what remains open.

3 Connected architecture and source provenance

One decoded frame feeds one SG1 detector. SG2 and SG3 share its landmarks and capture identity; SG4 then aggregates synchronized evidence before SG5 decides. The old fixed-FPS SG4 notebook was not used as the current integration target.

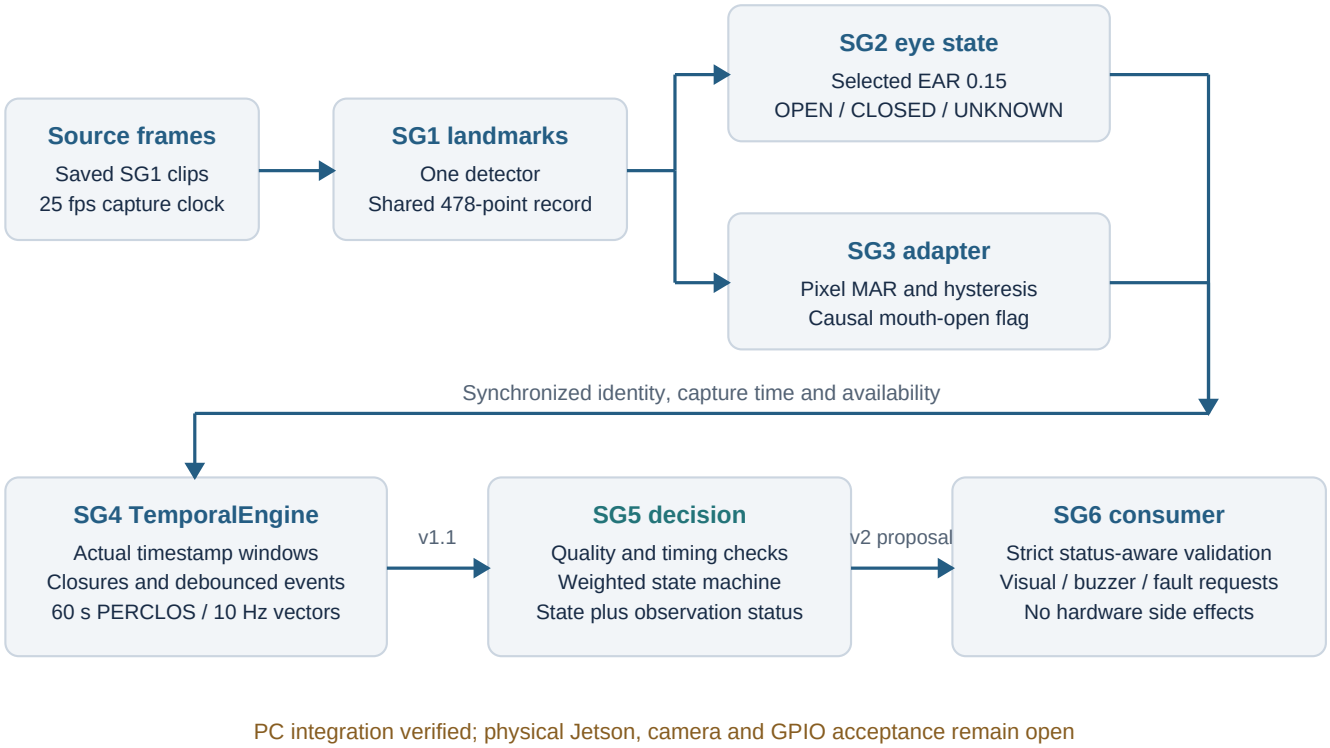


Figure 1. Implemented PC data path. SG3 and the availability gate are explicit adaptations; the final consumer emits hardware requests rather than operating a buzzer.

Owner	Source branch	Commit prefix	Reused responsibility
SG1	sg1	`5fcb93627efb`	Actual face/landmark detector and validator
SG2	sg2-midterm-integration	`ca73a39f72b2`	Selected EAR processor and cue adapter
SG3	sg3	`e3661b7575eb`	Reference MAR indices and thresholds; new causal adapter
SG4	saman	`1019abe55a5f`	Actual timestamp-based TemporalEngine and goldens
SG5	sg5	`9e0de1750fe3`	Preserved Week 5 A/B/C reference algorithms
SG6	sg6-week5-integration	`b4bf9ec42c8b`	Legacy validator and reported target evidence

Full commits, original Git-blob SHA-256 hashes, import scopes and attribution are versioned in `integration/sg5_midterm_sources.json`. The current SG4 source is branch **saman**. Source import is not an assertion of human review, acceptance or a main-branch merge.

4 Producer contracts and explicit adaptations

SG5 consumes SG4's current six-field v1.1 record without renaming its units or changing its temporal windows. All five native producer streams retain exact golden-output parity.

SG4 field	Interpretation
time_s	Increasing capture-clock seconds within one driver session
face_valid	Current face observation availability, 0 or 1
perclos	Valid eye-closed-time fraction, 0-1, in a 60-second window; not percent
eye_closure_s	Current ongoing eye-closure duration in seconds; zero when open/invalid
long_blinks	Closure-event count for durations at least 0.5 s, over 30 seconds
yawn_count	Debounced yawn-event count over 60 seconds, not yawn-frame count

SG2 hard states are not probabilities

The selected SG2 processor returns OPEN, CLOSED or UNKNOWN with explicit eye validity. Its closure probability and SG1's per-face detection confidence can be null. The narrow availability synchronizer preserves those absences instead of replacing them with thresholds or fabricated probabilities. SG4's actual window/event logic is retained. The first observation after a gap greater than 200 ms is censored in the annotated SG5 path, for both native and hard-state inputs, to break unsupported runs. Unannotated native replay retains reference behavior.

SG3 is a causal adaptation of the teammate experiment

The source Colab script includes notebook commands and offline future-interval analysis; it cannot be imported as a live module unchanged. The portable adapter reuses mouth indices 61, 291, 13, 14, 82, 87, 312 and 317. Coordinates are converted to pixels before distances are taken:

```
MAR = mean(d(13,14), d(82,87), d(312,317)) / d(61,291)
pixel_x = normalized_x * image_width
pixel_y = normalized_y * image_height
```

Inherited opening/hold thresholds are 0.50/0.45; confirmation requires 1.5 s. The raw active mouth-opening flag goes to SG4, which applies persistence/debounce once. The adapter's confirmed/event fields are diagnostics, not a second persistence input. Missing faces, invalid geometry or gaps reset continuity. Threshold calibration and owner acceptance remain open.

Quality is a separate observation property

Direct integration carries current eye validity in addition to SG4's historical validity fraction. A visible face with unavailable eyes is not a usable current eye observation. These are SG5-side metadata, not silently added fields in the frozen SG4 wire schema. The measured CLOSED fraction is not independently validated P80 eyelid occlusion.

5 Decision logic and observation status

The core is rule-based and imports only the Python standard library. No SG5 model is trained. The primary C design combines normalized evidence, an exponential smoother, hysteresis and timestamp-based confirmation. State and observation status remain distinct.

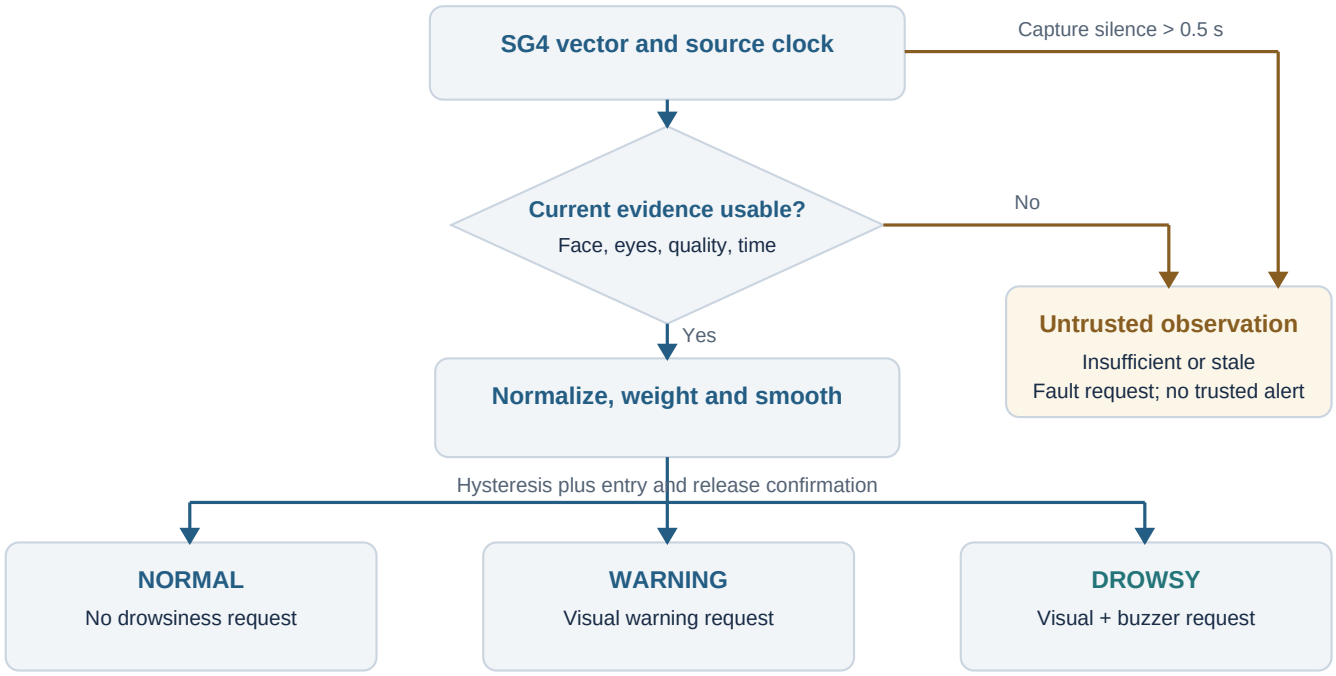


Figure 2. Driver state is trusted only when status is OK. An insufficient or stale observation retains an untrusted state, suppresses a trusted drowsiness action and requests a sensor fault.

```
z_i = clip((x_i - low_i) / (high_i - low_i), 0, 1)
S = 0.45*z_perclos + 0.20*z_closure
  + 0.20*z_long_blinks + 0.15*z_yawns
M_t = M_previous + (1 - exp(-dt / 2.0)) * (S_t - M_previous)
```

Normalization bounds (low, high): PERCLOS (0.05, 0.30), closure seconds (0.40, 2.00), long-blink count (1, 6), and yawn count (0, 3). Values outside each range are clipped.

Parameter	Stable C default	Purpose
WARNING entry / exit	0.20 / 0.14	Separate boundaries reduce threshold chatter
DROWSY entry / exit	0.65 / 0.53	Different entry and release requirements
Smoothing / entry confirmation	2 s / 1 s	Filter brief score changes; confirm the same higher target
Lower-risk confirmation	3 s	Require continuous lower evidence at each release boundary
Closure override	1.5 s	A valid prolonged closure bypasses score confirmation
Quality floor / source gap	0.55 / 0.5 s	Reject unusable evidence and restart temporal support

There is no forced five-second NORMAL warm-up in the new core. Invalid observations reset smoothing/confirmation; new sessions explicitly reset state. Duplicate, backward and nonfinite timestamps are rejected. None of the engineering score weights is a learned probability.

6 Controlled comparison on fixed development data

All five configurations receive identical six-field inputs from the two preserved synthetic drives. The allowlist excludes ground-truth labels from every decision call. The original A/B/C functions are extracted without GUI imports and remain unchanged; exact prediction parity passes for 16,200 values across six drive-method comparisons.

Development sequence	Rows	Duration	Role
Drive 1	3,000	300 s	Gradual drowsiness, short closure episode and recovery
Drive 2	2,400	240 s	Talking-related cues, tiredness, noise and face loss

Configuration	Accuracy	Macro F1	Events caught	Unmatched onsets	False D time s	Driver changes	Delay s
A reference	71.4%	0.698	2/2	2	1.1	22	8.35
B reference	75.2%	0.704	1/2	0	5.4	18	15.30
C reference	65.8%	0.641	2/2	0	8.6	11	11.60
C robust	64.1%	0.635	2/2	0	8.7	11	11.60
C fast recovery	64.0%	0.639	2/2	0	0.0	11	11.60

Pooled metrics weight samples and matched events rather than averaging per-drive percentages. Unavailable outputs remain UNKNOWN errors in the denominator. Rates are synthetic-development results, not real-driver accuracy. The raw driver-state changes above exclude status transitions.

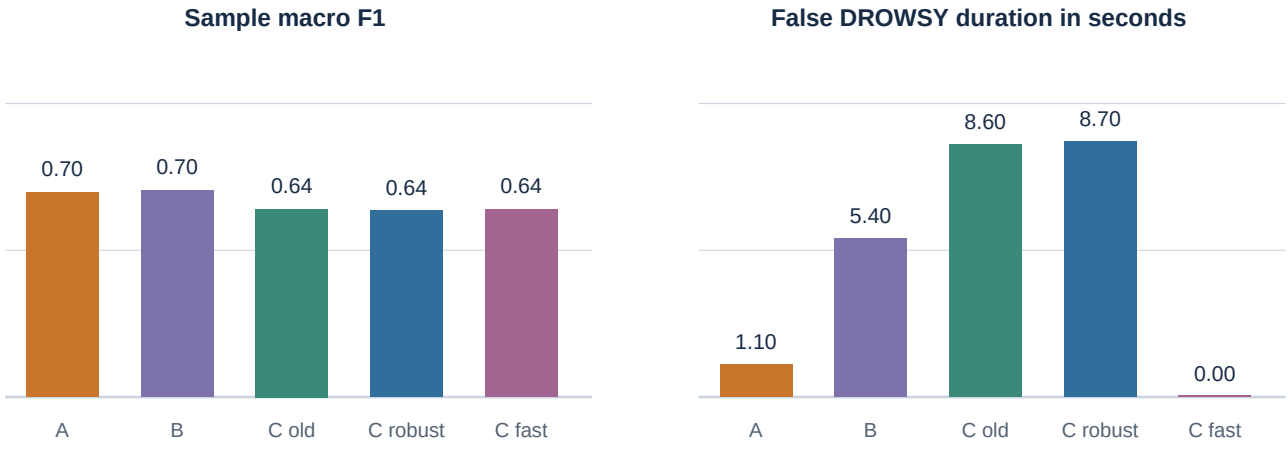


Figure 3. Higher macro F1 and lower false-DROWSY duration favor different configurations; the faster candidate is a sensitivity study, not an independently selected winner.

The actual SG4 implementation uses a **60-second** PERCLOS window; these legacy synthetic drives use **30 seconds**. The identical field name does not establish matching input distributions. Real SG4 thresholds must be re-evaluated on independent labelled recordings.

7 Failure analysis and selection rationale

Primary: robust C. Backup: threshold A. C is retained for explicit observation quality, correct time/session handling and interpretable state transitions, not as an accuracy improvement. A is a low-complexity, transparent fallback. B remains an implemented comparison and has the highest pooled sample accuracy on this development set.

Finding	Evidence	Engineering implication
B misses the short episode	One of two episodes detected; eye closure alone contributes at most 0.20 to a 0.50 DROWSY threshold.	Keep an explicit prolonged-closure rule rather than relying only on the blended score.
A fragments a longer episode	22 native state changes; strict one-to-one evaluation leaves a repeated onset unmatched.	Threshold crossings need temporal confirmation and a documented alert repetition policy.
C delays gradual detection and recovery	Strict mean delay 11.60 s for robust C; false-DROWSY overhang 8.7 s.	Collect labelled real streams; tune smoothing and release against latency and missed-duration costs.
Faster recovery trades recall for overhang	False-DROWSY time falls to 0.0 s but DROWSY sample recall drops.	Do not choose a winner solely because one development metric improves.

Robust C sample performance

Reference class	Support	Precision	Recall	F1
NORMAL	3631	0.852	0.647	0.735
WARNING	1000	0.305	0.577	0.399
DROWSY	769	0.861	0.698	0.771

Strict metrics are not interchangeable with the old scores

A detected event requires a **new predicted DROWSY onset inside its labelled half-open interval**, one-to-one. Preexisting overlap is recorded separately as coverage; it does not create a negative latency. Late onsets after unused episodes are diagnostic only. No detected event means null delay, not zero. False-DROWSY duration counts time spent DROWSY outside the labelled DROWSY state.

The original evaluator allowed many-to-one overlap and a five-second post-episode grace. A's two unmatched onsets here include one repeated onset and one late onset after an already detected episode. They are not evidence that A's unchanged algorithm suddenly produced two unrelated false incidents.

Robust C has **11 native driver-state changes and 16 availability-status changes**; the evaluated prediction trace therefore changes 27 times when non-OK status becomes UNKNOWN. It is misleading to label all 27 changes as driver-state flicker. Macro F1 is averaged over three target classes; UNKNOWN remains a prediction column and an error, not a fourth ground-truth class.

8 Connected execution on five saved videos

The complete PC path processed **2,932 source frames and 1,173 SG4 vectors**. All 1,173 decisions had status OK and passed the usable legacy structural adapter. The output contained 461 NORMAL and 712 WARNING predictions; no DROWSY prediction occurred. These counts are observations of the software, not validated driver states.

SG1 source	Frames	Vectors	Mean process ms	p95 process ms	Offline fps
Clip 01	720	288	12.78	13.57	34.64
Clip 02	577	231	12.75	13.57	34.19
Clip 03	560	224	12.79	13.64	34.12
Clip 04	475	190	12.90	14.18	34.44
Clip 05	600	240	12.96	14.12	34.53

Processing time excludes acquisition and annotation/encoding. Offline throughput includes file I/O. Neither is a live-camera or Jetson FPS measurement. The five runs used Windows Python 3.11.0, NumPy 1.26.4, OpenCV 4.11.0 and MediaPipe 0.10.21; SG5 was evaluated approximately at 10 Hz.

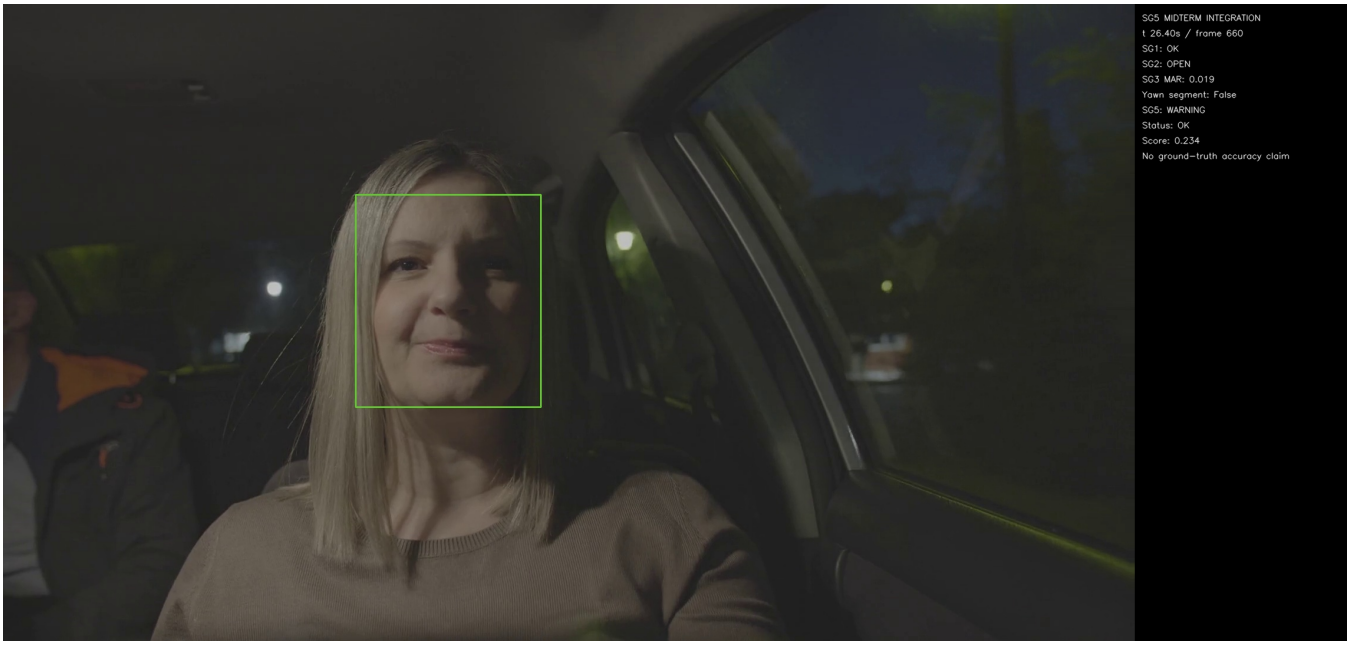


Figure 4. Saved Clip 01 observation at 26.40 s: actual SG1 face geometry, SG2 OPEN, SG3 MAR 0.019 and SG5 WARNING with status OK. A state can reflect historical cues even when the mouth is currently closed. The screenshot is an output example, not a ground-truth label.

What this run establishes

The source-to-consumer path executes without inventing confidence values or dropping source identity. Per-stage JSONL, source/model hashes, timings, example packets and annotated MP4s are available under each `results/videoNN` directory. Full source clips originate in SG1's branch.

Independent drowsiness, yawn and eye-event annotations were unavailable for these clips. No real-video precision, recall or accuracy is claimed. The SG3 segment counts are predictions. No DROWSY output here does not prove that all subjects were awake or that the system has high recall.

9 SG6 software handoff and fault behavior

The proposed `sg5.sg6.v2` packet has fourteen fields, strict types and explicit status. SG6's dry-run consumer validates each packet and produces idempotent requests. It has no camera, GPIO, network or sound side effects.

Packet fields	Meaning
schema_version; time_s; session_id	Version and source capture identity
driver_state; status	Retained NORMAL/WARNING/DROWSY, usable only while status is OK
score; smoothed_score; quality	Engineering evidence and observed quality, not trained probabilities
alert; alert_level; alert_event	Actionable state, level 0/1/2, and one-shot upward transition
transition; reason; config_id	Native state change, explanation and versioned rules

Actual saved prolonged-closure decision


```
{
  "schema_version": "sg5.sg6.v2",
  "time_s": 75.5,
  "session_id": "mock-drowsy",
  "driver_state": "DROWSY",
  "alert": true,
  "alert_level": 2,
  "score": 0.55838,
  "smoothed_score": 0.39804774974700563,
  "quality": 1.0,
  "status": "OK",
  "reason": "prolonged eye-closure override",
  "transition": true,
  "alert_event": true,
  "config_id": "midterm_c_v2"
}
```


Condition	Hardware request
OK + WARNING	visual_warning = true; no DROWSY buzzer request
OK + DROWSY	visual_warning = true; buzzer_requested = true
INSUFFICIENT_EVIDENCE or STALE	sensor_fault = true; no trusted drowsiness request

The exact legacy adapter returns only driver_state, alert and confidence for usable decisions; legacy confidence means observation quality. It raises for insufficient/stale observations. SG6's actual old validator accepts this shape. That proves structural compatibility, not acceptance of a new semantic policy. SG6 still owns alarm repetition, fault indication and physical wiring.

10 Verification and reproducible demonstration

Run from the repository root using the isolated Windows Python 3.11 environment. The decision core and consumer are standard-library-only; real SG4 replay needs the core requirements. The full vision path uses the validated PC pins. It does not require a Torch/CNN checkpoint in selected EAR mode.

```
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r module_5/midterm/requirements-pc.txt
.\.venv\Scripts\python.exe module_1/scripts/download_model.py
```

Create the environment only if absent. The SG1 downloader checks the official model SHA-256. This uses a pretrained landmark model [5]; neither landmark training nor SG5 model training is performed.

```
$py = ".\.venv\Scripts\python.exe"
& $py -m module_5.midterm.demo --scenario drowsy
& $py -m module_5.midterm.demo --scenario face_lost
& $py -m module_5.midterm.evaluate --plot
& $py -m unittest discover -s module_5/midterm/tests -v
& $py -m module_5.midterm.pipeline `
  --video module_1/data/videos/driver_sample_01.mp4 `
  --output module_5/midterm/results/video01 --annotated
```

Use the full output/consumer command sequence in `module_5/midterm/scripts.md`. The mock demo executes the actual SG4 engine; the vision command executes actual upstream inference. Those are intentionally separate demonstrations. Plots are headless and saved as PNGs.

Verification layer	What the check establishes
SG5 decisions and contracts	State timing, target-specific confirmation, invalid quality, current-eye availability, session/gap reset, strict schemas, legacy compatibility and silent delivery faults
SG3 geometry and temporal logic	Pixel MAR on rectangular frames, causal opening/hold behavior, one diagnostic event per confirmation, identity/time preservation and invalid-input reset
SG4 conformance	All five producer goldens match; actual SG4 vectors pass the v1.1 contract and run through SG5/SG6
Original baseline and metrics	Exact A/B/C parity, no label leakage, one-to-one episode matching, nonnegative latency, explicit UNKNOWN errors and separate status changes
Upstream regression	SG1/SG4 and SG2 tests are executed separately in compatible environments; upstream code remains attributed and unchanged

Recorded software verification: 255 passing tests across the SG5, SG1/SG4 and SG2 suites (96, 114, 45, respectively). SG2 regression uses the separate existing Torch environment; it is not a universal vision-environment claim. Dependency consistency also passed. Machine-readable commands, outputs and run evidence are recorded in `module_5/midterm/results/verification.json`. These are software checks, not target-hardware validation.

Producer tick uses the capture clock; transport silence uses SG6's independent monotonic-arrival watchdog. The live hardware loop must poll while no messages arrive. A file/stdin replay is not itself a continuously polling hardware watchdog.

11 GitHub governance and embedded readiness

Shared repository: Sananismm/CV_Drowsinees_Detection
Integration branch: sg5-midterm-integration
Team board: Computer-Vision Driver Drowsiness Monitoring, Project 2 (private; team access required).

Review request: https://github.com/Sananismm/CV_Drowsinees_Detection/pull/21. Opening a review request does not establish reviewed acceptance or a main merge.

The first Linux GitHub Actions software check passed for implementation commit `8ab3e2334009`. CI verifies software checks, not the physical target.

Tracked work	Evidence and next decision
Issue 8	Named warning-threshold / 1.5-second recovery sensitivity candidate and common-data comparison; review the trade-offs before closing.
Issue 9	Current SG4 implementation and native schema/goldens are used; connected PC runs exist. New adapter semantics still need owner review.
Issue 19	Midterm report, reproducibility and test evidence; link the recorded run and review request rather than marking completion from file creation alone.
Issue 20	SG3/SG4/SG6 acceptance and target-hardware checks remain distinct unresolved work.

The existing 27 October SG4 alignment note is retained without inventing SG5 attendance or new approval. The coordination register records producer/consumer responsibilities and unresolved decisions. Imported algorithm work is credited to its original subgroup; both SG5 members must explain their genuine contribution rather than treating assigned ownership as proof of authorship.

Embedded platform evidence has a separate boundary

Stage 1 C criterion	Current evidence and remaining action
Boot and environment 2 marks	SG6 must show actual Jetson Orin Nano 8GB boot, storage and OS. This preparation did not access the device.
Development stack 1 mark	PC pins and standard-library core are reproducible. Confirm ARM64 runtime compatibility, detailed landmarks and required software on the actual target.
Camera readiness 2 marks	Saved videos verify the software path only. SG6 must demonstrate USB/CSI capture or document the actual acquisition blocker.

SG6's inherited detector benchmark **reports** Haar 31.65 ms and YuNet 17.89 ms mean inference on 150 private recorded frames. It was not rerun here, excludes acquisition and SG2-SG5, and cannot be used as this pipeline's Jetson result. YuNet's five keypoints do not supply SG2's detailed eye landmarks. Physical camera, GPIO, target latency and memory remain unverified.

```
& $py embedded_6/sg5_handoff/check_platform.py `
  --output module_5/midterm/results/platform_pc.json
```

The recorded PC probe identifies Windows, not Jetson, and opens no camera by default. SG6 can run the same collector on its device, explicitly choosing --camera for a physical capture test. A video fallback cannot prove a connected camera.

12 Remaining work and references

Priority	Owner action and completion evidence
Interface acceptance	SG3 reviews causal pixel MAR and thresholds; SG4 reviews hard-state availability/gap behavior; SG6 reviews v2 status and fault requests. Record explicit decisions and tests.
Independent validation	Collect independently labelled real SG4 streams. Split development from evaluation subjects/clips; compare event recall, false duration, delay and coverage with the same protocol.
Target bring-up	SG6 records boot, exact runtime/model versions, camera capture and the hardware policy. Test no face, stopped producer and broken delivery before enabling physical alerts.
Profiling and next gate	Measure end-to-end acquisition/decision latency, throughput and memory on Jetson. Treat the present PC numbers as development measurements only.

Known limitations

The two synthetic drives are a small development set, and their PERCLOS window differs from the real producer. Five unlabelled video runs establish execution, not drowsiness accuracy or safe real-driving operation. MAR and EAR thresholds remain person/pose/lighting-sensitive. No independent P80 occlusion measurement, calibrated drowsiness probability or head-nod detector is claimed. A stopped or invalid sensor cannot establish an awake driver. Human interface acceptance and physical target validation remain required even when PC tests pass.

References and evidence locations

[1] CS477. *Midterm Project Evaluation Stage 1*, Week 8, pp. 1-2. Assessment authority for the current A15/B10/C5 scope and evidence boundary.

[2] CS477. *SEECS CV Face-Off 2026 Semester Project Guide*, pp. 3-7. SG5 role, interface-first development, accountability and Lab 8/9/11 integration gates.

[3] CS477. *Week 5 Subgroup Tasks GitHub Readiness Assessment*, pp. 1-3 and 6. Preserved baseline, controlled alternatives, common metrics and separate decision/UI responsibility.

[4] CS477. *Semester Project Topic Sub Group Allocations Section A B*, p. 2, rows 48-49. G1-D2 SG5 membership and student identifiers.

[5] Google AI Edge. *Face landmark detection guide for Python*. Official MediaPipe documentation, accessed 28 October 2026. Pretrained landmark inference and timestamped video API, not an SG5 training claim.

[6] Team repository snapshots and full hashes: `integration/sg5_midterm_sources.json`. Controlled evidence: `module_5/midterm/results/evaluation/evaluation.json`. Connected evidence: `module_5/midterm/results/video01` through `video05`, each with its own summary, source/model hashes and stream outputs.

Handoff conclusion. SG5 has a reproducible PC integration and a strictly validated software consumer boundary for SG6 review. The report does not substitute for owner approval, independent real-driver evaluation or the team's physical embedded-readiness demonstration.